STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**)**	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service	**)**	**Case 19-G-0310**

DIRECT TESTIMONY

OF

FERIT UCAR

ON BEHALF OF

ENVIRONMENTAL DEFENSE FUND

Dated: August 30, 2019

I. Introduction and Qualifications

Q. Please state your name and business address.

A. My name is Ferit Ucar. My business address is 257 Park Avenue South, 17th Floor, New York, New York, 10010.

Q. What is your occupation and by whom are you employed?

A. I am Senior Economist, Research & Analytics, at the Environmental Defense Fund ("EDF").

Q. Please describe your responsibilities in that position.

A. As a member of the Research and Analytics team in EDF's Energy Program, I conduct research and analysis to inform EDF's strategy and support its advocacy. My work focuses on modeling the economic, environmental, and public health impacts of energy policies, identification of transition pathways to a clean energy economy, and levers to transform the energy sector, with a primary focus on business model reforms in the electric and natural gas sectors. Since 2015, I have participated in the New York Public Service Commission's ("Commission") Reforming the Energy Vision proceeding, representing EDF's interests on matters related to the value of distributed energy resources, Earnings Adjustment Mechanisms ("EAMs"), and energy efficiency.

Q. Please describe your professional background.

A. I have over a decade of experience in researching and evaluating clean energy and energy efficiency programs. Prior to joining EDF, I consulted for the federal government, state agencies, and electric and gas utilities on energy efficiency and

energy affordability program design, implementation, and evaluation. In particular, I worked with the federal Low Income Home Energy Assistance Program ("LIHEAP") office to develop metrics and data to measure, track, and improve LIHEAP program performance.

Q. Briefly state your educational background.

A. I hold a Ph.D. degree in Economics from Princeton University and a Bachelor's degree in Mathematics and a Bachelor's degree in Economics from Koc University.

Q. Have you previously filed testimony before the New York Public Service Commission?

A. Yes. As a witness for EDF in the Consolidated Edison Company of New York, Inc. ("Consolidated Edison") rate case currently pending before the Commission (Cases 19-E-0065 and 19-G-0066), I provided pre-filed testimony regarding outcome-based EAMs.

Q. On whose behalf are you testifying in this proceeding?

A. I am submitting testimony on behalf of EDF.

II. Purpose of Testimony and Recommendations

Q. What is the purpose of your testimony?

A. The purpose of my testimony is to make recommendations concerning how The Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Gas East Corporation d/b/a National Grid (collectively, "Companies") can meet the evidentiary burden that the Commission will need to require in order to approve a rate plan after the effective date of the New York Climate Leadership and

Community Protection Act ("CLCPA" or the "Act"). In addition, my recommendations provide a structural approach whereby the Companies can quantify and potentially receive credit for successful outcomes from their efforts to reduce emissions under the Act.

Q. Please provide a summary of your testimony and recommendations.

A. My testimony describes the greenhouse gas ("GHG") emission reduction goals that have been adopted by New York State through the CLCPA; one of the mechanisms through which agencies such as the Commission are required to ensure that the State remains on track to achieve those goals; and the evidentiary showing that utilities, when they are seeking to obtain the Commission's approval of a rate plan, will need to make in order for the Commission to make the determinations of consistency and non-interference with the CLCPA that the Act requires in connection with any such approval. It also describes progress that has been made in a different setting, the EAM Collaborative arising out of Consolidated Edison's 2016 rate case, in considering what information is relevant for a similar analysis and what obstacles need to be overcome in order for that information to be obtained in a timely fashion. Finally, I describe potential opportunities for the Companies to reap benefits from exercising climate leadership in developing accurate emissions accounting tools in parallel with their leading GHG reduction activities.

Q. Was your testimony prepared by you or under your supervision?

A Yes.

Q. Are you sponsoring any exhibits?

A. Yes, I am sponsoring the following exhibits:

Exhibit __ (FU-1): CV of Ferit Ucar

III. New York's Climate Goals

Q. Please describe the policies relevant to GHG emissions that were in effect in the Companies' service territories and relevant to the Companies' operations at the time that the Companies filed their initial rate case on April 30, 2019 and corrections and updates on July 3, 2019.

A. As the Companies' Future of Heat Panel noted in its testimony, prior to the filing of the rate case, Governor Cuomo had adopted goals including 40% GHG emissions reductions (from 1990 levels) from the energy sector by 2030, the longer-term goal of decreasing total carbon emissions 80% by 2050, as well as a requirement that 100% of electricity consumed in New York be carbon neutral by 2040.[1] In addition, in May 2017 New York adopted a Methane Reduction Plan, which identifies 25 actions that are expected to "reduce methane emissions in alignment with New York State's commitment to reduce greenhouse gas emissions"[2] and, among other things, calls upon the New York Department of Environmental Conservation ("DEC") and the New York Department of Public Service ("DPS") to "[e]valuate and identify best technology and methods to identify leaks in each portion of the

[1] Future of Heat Panel testimony at page 7.

[2] N.Y. Methane Reduction Plan at 4 (May 2017), https://www.dec.ny.gov/docs/administration_pdf/mrpfinal.pdf.

system."[3] Further, in April 2019, shortly before the filing of the rate cases, the City of New York passed Intro 1253, which requires buildings over 25,000 square feet to meet building emissions limits that are in alignment with the City's own goals of 40% reductions by 2030 and 80% reductions by 2050 (from 2005 levels). This new City law, which is now known as Local Law 97, applies to the Companies' customers in the portion of the Companies' service territories that is in New York City.[4]

Q. Have any significant developments involving New York State climate policy occurred since the Companies' filed their initial rate cases on April 30, 2019 and corrections and updates on July 3, 2019?

A. Yes. On July 18, 2019, Governor Cuomo signed into law the Climate Leadership and Community Protection Act.[5]

Q. Please provide an overview of the key provisions of the Act.

A. The Act requires New York to adopt measures to reduce statewide GHG emissions by 85 percent by 2050 from 1990 levels, with an additional goal of achieving net zero emissions across all sectors of the economy by that time (the remaining 15 percent can come from carbon offsets).[6] The Act ensures that the short-term climate impact of methane, which causes 84 times as much global warming as the same

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3 *Id.* at 5.

4 New York City Local Law 97 (2019).

5 New York State Climate Leadership and Community Protection Act (CLCPA), https://legislation.nysenate.gov/pdf/bills/2019/S6599.

6 CLCPA, https://legislation.nysenate.gov/pdf/bills/2019/S6599.

amount of carbon dioxide over a twenty-year horizon,[7] will be recognized in GHG reporting.[8] The Act also recognizes that GHG emissions attributable to New York include "greenhouse gases produced outside of the state that are associated with the generation of electricity imported into the state and the extraction and transmission of fossil fuels imported into the state,"[9] and seeks to ensure that GHG emissions reductions are "real, permanent, quantifiable, verifiable, and enforceable."[10]

IV. CLCPA Implementation

Q. Does the Act define a role for the Commission in ensuring that the State meets its ambitious GHG limits?

A. Yes. The Act establishes a new Section 66-p of the Public Service Law, which requires the Commission to establish a program to require that (a) a minimum of 70% of the statewide electric generation secured by jurisdictional load serving entities to meet the electrical energy requirements of all end-use customers in New York State in 2030 be generated by renewable energy systems, and (b) that by the year 2040, 100% of the statewide electrical demand be met with carbon-free generation resources. The new Section 66-p will further require that, beginning in 2024 and again every two years thereafter the Commission conduct a

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7 IPCC, 2014: Climate Change 2014: Synthesis Report. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. *See* Box 3.2, Table 1.

8 Environmental Conservation Law § 75-0101(2) and §75-0105.

9 Environmental Conservation Law § 75-0101(13).

10 Environmental Conservation Law § 75-0109(3)(b).

comprehensive review of these programs, including progress toward meeting targets. Section 66-p further provides that, no later than July 1, 2024, the Commission shall establish programs to require procurement of at least 9 gigawatts (GW) of offshore wind electricity by 2035 and 6 GW of photovoltaic solar generation by 2035, and to support 3 GW of statewide storage capacity by 2030. Section 66-p also directs the Commission to specifically provide a minimum level of benefit for low-income customers and disadvantaged communities. In addition to these specific provisions applicable to the Commission, the Act provides that in issuing approvals and decisions, *all* State agencies, offices, authorities, and divisions "shall consider whether such decisions are inconsistent with or will interfere with the attainment of the statewide greenhouse gas emissions limits established in Article 75 of the Environmental Conservation Law…."[11]

Q. What information would be necessary for the Commission to assess whether approving utility proposals would be inconsistent with and/or would interfere with compliance with statewide GHG emissions limits?

A. First of all, the Commission would need a comprehensive current baseline of GHG emissions in each regulated utility service territory. Once the baseline is established, the Commission would need information and data to be able to assess actual progress being made in each utility service territory towards GHG emissions reductions, ideally on an annual basis to assess consistency with the Act. Finally, the Commission would need to be provided with estimated GHG impacts from the

[11] CLCPA, Section 7(2).

utility proposals as well as projections of GHG emissions with an assessment of variability and uncertainty to be able evaluate whether proposals interfere with compliance with future statewide GHG emissions limits as required by the Act.

V. Role of Utilities in facilitating CLCPA Implementation

Q. Do you have any recommendations regarding how the Commission should make a determination of consistency with and non-interference with State GHG goals?

A. The Commission's ability to make this determination is premised upon receiving sufficient information and analyses from the proponent of the request. In the case of a utility rate case, the proponent would generally be the utility.

Q. Are utilities well positioned to play a role in CLCPA implementation?

A. Yes. The decisions of utilities are central to the achievability of the State's climate goals. In fact, in this case, the Companies acknowledged their role in achieving the then-applicable State climate goals, citing to the State's goal of decreasing carbon emissions 80 percent by 2050, and National Grid's "Northeast 80x50 Pathway." The Companies state that their "gas networks will play an integral role in meeting these ambitious goals and delivering the lower-carbon economy of the future."[12]

Q. In your opinion, have gas utilities to date provided sufficient means and tools to make such a demonstration?

[12] Future of Heat Panel testimony at page 7.

A. No. As my colleague, Naim Jonathan Peress, has previously asserted to the Commission, gas utilities have generally not provided the tools or the means to assess and weigh climate impacts—good, bad, or indifferent.[13]

Q. What information or tools should regulated utilities provide to assist the Commission in assessing consistency with and non-interference with CLCPA?

A. Regulated utilities will need to develop a comprehensive GHG assessment methodology, provide the data, do the monitoring and knowledge management needed to organize, update, and augment the data, and then plug in the data in empirical tools and metrics to assess the GHG impacts of their operations. Ideally, these data and tools could be used to inform utility decision making such as around supply arrangements and infrastructure investment as well as to assist the Commission in making the determinations required under the Act. The tools that are used to assess current and forecasted GHG emissions should make use of the "best available economic models, emissions estimation techniques and other scientific methods," as contemplated by sections of the Act addressing methodological considerations.[14] These tools could be used to establish a current baseline, and after such baseline is established, measurement of GHG emissions in subsequent years would allow assessments of progress toward GHG goals and the

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[13] *In the Matter of Staff Investigation into a Moratorium on New Natural Gas Services in the Consolidated Edison Company of New York, Inc. Service Territory*, Case No. 19-G0080, Public Statement Hearing Transcript at page 128 (February 13, 2019) (statement of N. Jonathan Peress on behalf of EDF).

[14] *See* Environmental Conservation Law, § 75-0103(14)(b).

accuracy of projections, and to inform any adjustments needed for future projections of GHG emissions.

Q. Can you describe the appropriate scope and content of a GHG assessment tool that would provide the Commission with the information it would need to determine consistency with and non-interference with the CLCPA?

A. An appropriate GHG assessment methodology would be a holistic analysis of GHG emissions within a service territory, taking into consideration all emissions (or at a minimum all energy-related emissions) within the geographical footprint of the service territory, not solely those emissions that are directly attributable to the utility's own system. Such a holistic methodology would be appropriate primarily because of the nature of the GHG emissions limits established in the Act, which pertain to economy-wide emissions, including upstream emissions.

Q. Is it unreasonable to judge a request by a utility that only owns and operates one energy system by such a holistic GHG metric?

A. No. It might be unreasonable to judge a utility's own climate impact by a holistic metric, but that is not the nature of this exercise. The purpose of this metric is to give the Commission the foundation it needs to find that approving the utility's proposal will not handicap achievement of statewide targets. No methodology focused solely on emissions attributable to a specific utility's system would allow the Commission to make such a determination as long-term statewide GHG outcomes arise as a result of many factors and actions of multiple actors, all of which are interdependent. For example, utility actions and practices affect the

actions and practices of customers, suppliers, and other parties, shaping market outcomes that do not arise directly from the utility's own system. Gas utility supply decisions may indirectly affect the level of GHG emissions associated with natural gas extraction taking place in various locations; their treatment of natural gas customers that are electric generators may affect the rapidity with which the electric generation fleet is converted to lower-emitting or non-emitting generation; and their practices with respect to line extensions and non-pipeline solutions may affect the speed of electrification of building heat. There are also second-order effects; if natural gas utilities influence the GHG emissions intensity of the electric generation fleet serving their service territory, they will also, indirectly, influence the GHG emissions intensity of newly electrified applications, such as transportation and electric building heat. Knowing that their rate plans will be evaluated on a holistic basis with respect to all these indirect market outcomes aligns the natural gas utilities' business goals with the state's emissions limits and reduces the effect of incentives they might otherwise face to engage in activities that could be expected to hinder achievement of the GHG emissions limits.

Q. As a practical matter, what steps do gas utilities need to take to enable this holistic evaluation?

A. Gas utilities need to work with state and local entities to gather the requisite, economy-wide information that will make a holistic GHG emissions assessment feasible. This will require measuring emissions associated with the Company's own operations, including greenhouse gas emissions resulting from the combustion of

natural gas the Company delivers to customers as well as an accurate assessment of methane loss from the Company's own distribution system. It will require the Company to obtain reliable information about the upstream emissions associated with the natural gas it delivers, as well as information about greenhouse gas emissions in other sectors of the economy. In its fully developed form, this may require gathering information from private parties that are not regulated by the Commission, such as fuel suppliers, and accounting for methane loss at customer premises. Using this economy-wide information, a current status of emissions attributable to the Company's geographical footprint can be established.

After establishing a current, holistic GHG baseline for its service territory, the utility would need to provide projections of GHG emissions for future years based on projected economy-wide energy consumption and pertinent emissions factors. Projections of energy consumption for each fuel should take into account the impacts of energy efficiency, electrification, population growth, and other relevant factors. Projections of emission factors for natural gas should take into account the estimated impacts from renewable natural gas ("RNG") and biogas and changes in methane leakage rate from the natural gas system (areas for which the Company proposes new projects during the rate plan to reduce its emissions). Projections of emissions factors for electricity should take into account the changing GHG emissions-intensity of the generation fleet, as required by the CLCPA. Moreover, since any future projections would contain inherent uncertainty and variability,

these projections should be accompanied by an uncertainty and sensitivity analyses in order to provide the Commission with necessary information to evaluate risks.

Q. Would the establishment of a baseline and a means to assess progress provide a useful operational tool for the Companies?

A. Yes. At the present time, the Companies have at their disposal various means of reducing GHG emissions in their geographical footprints. In fact, the Companies are proposing numerous initiatives to reduce emissions including: 1) investments to source and procure RNG through the Newtown Creek project with the City;[15] 2) a green gas tariff whereby customers can subscribe to receive a portion of their gas supply from RNG;[16] and 3) a power-to-gas demonstration project in partnership with the US Department of Energy and National Renewable Energy Lab.[17] These projects could lead to significant GHG reductions, which should be quantified. The sooner the Companies begin quantifying overall emissions and planning to support the dramatic reductions required under the Act, the more their business practices and the resulting market outcomes can be expected to align with the goals of Act. Comparatively early compliance with the goals of the Act would be advantageous for the Companies because they could reasonably expect the Commission to look more favorably upon the proposals of Companies that take such initiative, and, aided by such suitable tools, the Companies could work with the Commission to

[15] Future of Heat Panel testimony at page 76.

[16] *Id*. at page 33.

[17] *Id*. at page 84.

develop a system of incentives for emissions reductions that take place ahead of the schedule required under the Act. Conversely, the later the Companies establish a defensible baseline, the harder it will be for them to obtain full recognition for their early leading-edge work to reduce economy-wide GHG emissions.

Q. Would the establishment of incentives that reward regulated utilities for climate leadership where GHG reductions are achieved faster than required by law be appropriate?

A. Yes. Because climate change is a result of the cumulative GHG emissions that are present in the atmosphere and there is limited time to reduce them before the point at which climate change becomes irreversible and catastrophic,[18] such earlier-than-mandated GHG emissions reductions may have more value than emissions reductions in later years. The New York's Legislature recognized the vital importance of climate leadership in the recitals to the Act, finding and declaring as follows:

> "Action undertaken by New York to reduce greenhouse emissions will have an impact on global greenhouse gas emissions and the rate of climate change. In addition, such action will encourage other jurisdictions to implement complementary greenhouse gas reduction strategies and provide an example of how such strategies can be implemented. It will also advance the development of green technologies and sustainable practices within the

[18] *See generally* 2018 IPCC Special Report, Global Warming of 1.5°C, *available at* http://www.ipcc.ch/report/sr15/.

private sector, which can have far-reaching impacts such as reduction in the cost of renewable energy components, and the creation of jobs and tax revenues in New York."[19]

The N.Y. Legislature further found and declared:

> "By exercising a global leadership role on greenhouse gas mitigation and climate change adaptation, New York will position its economy, technology centers, financial institutions, and businesses to benefit from national and international efforts to address climate change."[20]

In light of the potentially high value of early GHG reductions, the commensurately high cost of delay, and New York's stated intent to exercise leadership in the struggle to mitigate climate change, incentives for regulated companies that reward climate leadership would be appropriate.

Q. For the years 2014 through 2018, the Companies provide estimated methane emissions reductions associated with the replacement of leak prone pipe.[21] Do these estimates provide an accurate assessment of methane loss from the Company's system?

A. No. The Companies' emission reduction estimates are based on the methodology set forth in the Environmental Protection Agency's ("EPA") regulations.[22]

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[19] CLCPA, Section 1(3).

[20] CLCPA, Section 1(12).

[21] Future of Heat Panel testimony at pages 14-15.

[22] *Id.* at page 15 (citing 40 C.F.R. § 98, Subpart W).

However, recent academic findings have demonstrated that observed methane emissions are about twice that reported in the EPA inventory.[23] In practice, the only reliable way to get an accurate assessment of methane emissions is to measure them. The need for accurate GHG accounting in the context of the CLCPA provides another reason why the Companies should promptly integrate Advanced Leak Detection technology into their business practices system-wide. A comprehensive and accurate measurement of methane emissions is crucial to establishing an accurate GHG baseline for the Companies and tracking their progress toward reducing emissions.

Q. How would the creation and establishment of ongoing, comprehensive, and accurate measurement of methane emissions assist the Companies in efforts to meet the goals of the CLCPA?

A. As further described in the testimony of Joe von Fischer, such a measurement system can provide more accurate and useful tools in the Companies' leak prioritization efforts. Combining advanced leak detection technology with traditional leak surveys offers utilities unique insight into their systems that is not possible using only traditional leak survey methods. Deploying these methods will also assist the Companies in ensuring a more accurate accounting of GHG emissions, consistent with the Act.[24] Moreover, as stated previously, it is in the best

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[23] G. Plant *et al.*, Large Fugitive Methane Emissions from Urban Centers Along the US East Coast, https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2019GL082635.

[24] For example, both Companies provide a high level estimate of potential methane emissions reductions over the next several years based on each respective company's

interest of the Companies to identify and quantify the largest sources of methane emissions in their service territories now to set an accurate GHG baseline because detection of such large sources at a comparatively late date could cast doubt on earlier-demonstrated achievements of GHG reductions and hamper the Companies' ability to have future emissions trajectories deemed to be in compliance with the requirements of the Act.

VI. Prior Efforts to Develop this Holistic Perspective

Q. Is there any precedent for aligning a utility's business model with holistic GHG outcomes?

A. I am not aware of any full deployment of such an approach, but we have previously introduced a similar concept in the Consolidated Edison collaborative that followed that utility's 2016 rate case, where we proposed it as the basis of an EAM. In the collaborative, parties considered various ways to use an EAM to align Consolidated Edison's earnings opportunities with favorable GHG outcomes. Most of the discussion in the collaborative concerned a metric that would add up specifically identified ("targeted") emissions-reducing activities and reward Consolidated Edison for sufficient levels of such activities. On behalf of EDF, I encouraged parties to consider a "broad" metric, as an alternative to this targeted approach. In the EAM context, the term "broad" provides a useful contrast to "targeted," but other than when referring to the EAM opportunities considered in the Consolidated

current facilities assuming LPP retirement pursuant to each company's proposed program. Gas Infrastructure and Operations Panel, Case 19-G-0309 at page 30; Gas Infrastructure and Operations Panel, Case 19-G-0310 at page 29.

Edison collaborative, the term "holistic" provides a better description of the nature of the analysis that I am recommending.

Q. Please describe what you meant by a "broad" metric in the EAM context.

A. In the context of an EAM, a "broad" GHG metric would assess measured, holistic, economy-wide GHG emissions outcomes in the utility service territory and reward the utility for the achievement of GHG emissions reductions above and beyond the historic trend and/or business-as-usual projections. This would automatically recognize GHG reductions of all types, including emissions reductions that result from improved utilization of utility energy, emissions reductions that result from improved price signals, emissions reductions that result from reduced carbon intensity of the utility's energy resources themselves (for example, substituting RNG/biogas for conventional natural gas), emissions reductions that result from substituting usage of energy provided by a utility for more carbon-intensive practices not involving the utility at all (for example, electrification of vehicles, substituting natural gas for higher-emitting delivered fuels, or electrifying heating applications that would otherwise have been fossil fueled), as well as emissions reductions from non-energy sources (for example, capturing fugitive methane emissions from waste to produce biogas).

Q. Why would a "broad" metric be advantageous for evaluating greenhouse gas outcomes in an EAM?

A. While an EAM based on a "broad" metric would be more challenging than a targeted metric to design as the impact of exogenous factors like weather and the

state of the economy would have to be isolated – and the proper time horizon might be longer than a typical rate case settlement plan – a "broad" metric could overcome the various limitations associated with a narrowly-defined targeted GHG reductions metric. Compared to the targeted approach, a "broad" metric would be much simpler for the utility and its regulator to calculate because it would have fewer elements that would have to be calculated solely for purposes of computing the metric. The shortcomings of a narrow approach to the EAM are also relevant to my reasons for recommending a holistic, emissions-measurement-based assessment in connection with CLCPA consistency and non-interference. First, the narrow approach relies on "deemed" values and does not assess measured GHG emissions reductions, which could deviate materially from the values assumed before the fact. Additionally, such an approach fails to provide the utility with any incentive for ensuring that short-term actions are taken in a manner that is consistent with optimal long-term GHG performance. The narrow metric also misses GHG emissions reduction opportunities beyond the targeted set of activities, and does not account for interactive effects among activities, whether favorable or unfavorable with respect to emissions outcomes. It also fails to account for market effects (indirect effects that result from the activities that the utility is tracking) as well as any activities that are not necessarily known to the utility. For all these reasons, a metric based on a set of pre-selected activities, however carefully designed, in effect predetermines the utility's actions, and so fails to provide a significant incentive for

innovation, including innovation that is needed to achieve the requirements of the CLCPA.

Q. Did the Collaborative recommend adoption of a "broad" GHG metric as the basis of an EAM?

A. No, largely for reasons of feasibility. As the Commission observed in the order approving the metrics proposed by the Collaborative participants in a Report, "[t]he Report…endorses the Broad GHG Reduction metric as a scorecard metric without associated financial incentives. The Broad GHG Reduction metric measures GHG reduction in New York City without attributing those reductions to specific technologies or market activities in the Consolidated Edison service territory based on New York City's annual GHG Emissions Inventory. The Report notes that while the parties to the Collaborative agree that the Broad GHG Reduction metric has merit, this metric was not selected as an EAM for 2019 due to the complexities of developing both a service territory-wide inventory [footnote omitted] and targets to meaningfully measure achievement."[25]

Q. Please describe the complexities of developing a holistic, "broad" metric, which made it infeasible to deploy as an EAM in the short term.

A. The most immediate barriers were very concrete. Although a comprehensive GHG inventory exists for New York City, which is most of Consolidated Edison's service

[25] Case 16-E-0060, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Electric Service*, Order Approving Recommendations of the 2018 Outcome-Based Earnings Adjustment Mechanisms Collaborative Report (April 25, 2019) at 4-5.

territory, there is no such inventory for Westchester. Additionally, as the 2018 Report from the Consolidated Edison Collaborative noted, the New York City GHG Inventory, which could provide the backbone for such a metric, generally lags too far behind to be useful for an EAM on the typical EAM schedule. Other complexities would come into play when establishing targets and earnings opportunities for a holistic EAM. Compared to a targeted GHG emissions reduction EAM, a broad GHG emissions EAM would constitute a far more significant departure from conventional performance-based regulation, and would require thoughtful consideration of how utility practices can contribute to large-scale market outcomes. But in any event, a means to quantify and assess conformity with the Act is necessary, and developing those structures sooner (e.g., beginning immediately) will improve the Companies' ability to manage the full range of emissions reduction levers available to them and facilitate early emission reductions, advancing the goals of the Act.

Q. What is the current status of the "broad" metric as it relates to Consolidated Edison?

A. The 2018 Report of the EAM Collaborative contemplated that it would be reported as a Scorecard, and very basic data on GHG emissions from the stationary energy and transportation sectors in New York City for the Calendar Year 2017 were included in the 2018 EAM report, which was filed in April 2019.[26] However, I am

[26] Case 16-E-0060, *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for*

unaware of any efforts to fill methodological gaps that make it infeasible to include the portion of the service territory that is in Westchester or to coordinate with the City of New York to expedite the collection and availability of pertinent data to address the time lag.

Q. Is this holistic approach to greenhouse gas emissions appropriate where, as here, the utility under consideration is a gas-only utility, rather than a combination utility (as Consolidated Edison is in most of its service territory)?

A. Yes. First, the holistic approach accurately aligns with the exercise of giving the Commission the information it needs to determine whether the utility's proposed business plans are consistent with and will not interfere with achievement of the state's statutory GHG emissions limits. By contrast, any evidentiary showing limited to the operation of the natural gas system would provide little or no basis for making any such determination. Moreover, even gas utilities have an opportunity to contribute – favorably or unfavorably – to GHG emissions outcomes from sectors other than their own (including vehicular emissions and emissions from heating buildings that are currently heated using delivered fuels). A holistic evaluation of service territory emissions properly gives the utility a reason to be mindful of such opportunities to influence those other sectors when developing long-term plans, investing in infrastructure, and generally operating its business.

Electric Service, Consolidated Edison 2018 Energy Efficiency Earnings Adjustment Mechanism Achievement Report (April 1, 2019) at 11.

That said, it goes without saying that, compared with a combination utility, a gas-only utility has fewer levers to pull that affect economy-wide GHG emissions. That may be an important consideration when setting targeted levels of economy-wide GHG emissions reductions for purposes of providing a utility with an incentive in the form of additional earnings opportunity. However, that consideration is not particularly relevant where, as here, the question is not whether the utility should be entitled to additional earnings, but instead concerns consistency (or lack thereof) with State GHG emissions limits that are themselves not specific to the natural gas system.

Q. How does your prior thinking about the "broad metric," and the Consolidated Edison collaborative experience, inform your recommendations about the holistic approach to providing evidence for a determination of consistency and non-interference in the context of the CLCPA?

A. My prior experience suggests that although the reasons for taking a holistic approach are compelling, the practical barriers to doing so are substantial. There is no time to lose in developing detailed, accurate, and timely GHG emissions inventories for individual service territories, to assess consistency with goals when the relevant deadlines are upon us and to provide a current baseline and a holistic way to measure progress in reducing GHG emissions, and for the Companies to identify and implement practices that facilitate such progress and be recognized for their achievements in this area by their regulator. These inventories have to be performed on a recurring basis and cannot be excessively lagging. Although the

utilities themselves will need to provide this information to the Commission to support Commission approval of their plans, the utilities are not, presently, especially well situated to create economy-wide emissions inventories. They will need to work with local and state government entities to create these inventories as well as with electric utilities in their gas service area. The gas utilities should start with their systems and get accurate emissions inventories associated with delivery and consumption of natural gas in their service territory while beginning to work with local and state government entities and electric utilities on the other components of the inventory. The CLCPA directs the Department of Environmental Conservation to issue a statewide GHG emissions report within two years and each year thereafter.[27] The Act also provides guidance on how to develop GHG inventories for this report.[28] In developing GHG emissions data for their geographic footprints, utilities should follow the same guidance to ensure that their methodologies are consistent with those being used to develop the statewide greenhouse gas emissions inventory and report.

Q. Does this conclude your testimony?

A. Yes.

[27] *See* Environmental Conservation Law § 75-0105(1).

[28] *See* Environmental Conservation Law § 75-0105.

Exhibit __ (FU-1):
CV of Ferit Ucar

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PROFESSIONAL EXPERIENCE

ENVIRONMENTAL DEFENSE FUND, New York, NY 2015 – Present
Senior Economist, Research & Analytics, Energy Program

- Evaluate policy options and instrument choices to promote clean energy, reduce energy consumption, and cut greenhouse gas emissions.
- Engage in original research and in-depth economic analysis in support of clean energy policies.
- Build internal analytics team capacity to evaluate new research, studies, models, and data.
- Produce white papers, fact sheets, case studies and best practice manuals for economic policy options to help state advocacy leads meet their goals.
- Disseminate successful policies and initiatives for use in other states, federal legislative efforts and rulemaking, and internationally.
- Provide material to support testimony in public utility commission hearings, legislative sessions and Federal Energy Regulatory Commission.
- Summarize research for broad audience, as well as public communications including Congressional testimony, Op-Eds, articles, and blog posts.
- Identify and coordinate with key internal and external experts in economics, finance, engineering, policy, and environmental science fields.
- Develop analytical foundation for EDF's clean energy work, including metrics to measure and track progress toward strategic goals and key donor deliverables.
- Represent EDF at public meetings including policymakers, academic conferences and workshops, gatherings of advocacy organizations, and general public.

NMR GROUP, INC., Somerville, MA 2014-2015
Senior Quantitative Analyst

- Conducted research and evaluation of energy-efficiency and renewable energy programs.
- Developed macroeconomic models to estimate impact of ratepayer-funded residential and commercial energy efficiency programs.
- Designed complex sample designs and weighting schemes for survey research in order to obtain information of sufficient accuracy at lowest possible cost.

APPRISE, Princeton, NJ 2008-2013
Senior Project Director

- Designed and implemented research and evaluation projects on programs that deliver bill payment assistance, energy education, and energy efficiency services to low income households.
- Provided technical assistance to US Department of Health and Human Services in completing LIHEAP Report to Congress, assessing LIHEAP program performance, and developing energy information and data for grantees.
- Assessed consistency, sufficiency, accuracy, and efficiency of existing data systems for State LIHEAP offices and made recommendations for improvements.
- Conducted survey research, including sample frame development, sample design and selection, survey design and testing, data collection implementation, and analysis of survey data.
- Trained and supervised team of policy analysts, research associates, and database programmers.

PRINCETON UNIVERSITY, Princeton, NJ 2002-2008
Graduate Student and Teaching Assistant

SAGE PUBLICATIONS, London, UK 2000-2002
Editorial Assistant, International Journal of Cross Cultural Management

EDUCATION

Princeton University
Ph.D., Economics, 2008.

Koç University
B.S., Mathematics and B.A., Economics, 2002.

OTHER INFORMATION

PUBLICATIONS

- E. Stein and F. Ucar, *Driving Environmental Outcomes Through Utility Reform: Lessons from New York REV,* EDF WHITEPAPER (January 2018).
- C. Russell, et al., *The View from the Top: Top-down Estimation of Program Savings in Massachusetts,* INTERNATIONAL ENERGY PROGRAM EVALUATION CONFERENCE (August 2015).
- M. Blasnik, et al., *Evaluation of the Weatherization Assistance Program during Program Years 2009-2011 (American Recovery and Reinvestment Act Period): Energy Impacts for Single Family Homes*, OAK RIDGE NATIONAL LABORATORY, ORNL/TM-2014/582 (September 2014).
- B. Tonn, et al., *Weatherization Works-Summary of Findings from the Retrospective Evaluation of the US Department of Energy's Weatherization Assistance Program*, OAK RIDGE NATIONAL LABORATORY, ORNL/TM-2014/338 (September 2014).
- M. Blasnik, et al., *National Weatherization Assistance Program Impact Evaluation: Energy Impacts for Single Family Homes*, OAK RIDGE NATIONAL LABORATORY, ORNL/TM-2015/13 (September 2014).
- J. Berger and F. Ucar, *Comparison of Pooled and Household Level Usage Impact Analysis,* INTERNATIONAL ENERGY PROGRAM EVALUATION CONFERENCE (August 2013).
- F. Ucar, *Three Essays in Health and Labor Economics*, PHD DISSERTATION (September 2008).

HONORS AND AWARDS

- Towbes Prize for Outstanding Teaching, Princeton University, 2007.
- Werner von Siemens Excellence Award, Koç University, 2002.